Distributed control Framework of Multirobot system using Automation and Control Applications

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***Abstract*— In this paper, a disributed control framework of multirobot system will be described in which a SIMATIC S7-1500 PLC (programmable logic controller) is employed to provide a stable source of sequence control, interlocks and safety logic, and MATLAB is employed to provide a higher level of coordination, monitoring and processing of motion feedback. The process of data exchange in both directions between the PLC and MATLAB is done through OPC UA and operator control is done using WinCC HMI/SCADA with clear modes, alarms and diagnostics. Stable communication, proper work of the process sequence, and online reception of the results of movement of the robot during operation were experimentally tested and confirmed. The proposed solution helps to transform a set of manipulators into an operator-friendly industrial automation system with a focus on any high-tech system.**

I. INTRODUCTION

Robot manipulators have become an essential part of modern automation and are widely used in manufacturing and related industries due to their accuracy, repeatability, and ability to perform hazardous or repetitive tasks [2,9]. Industrial robotic systems are currently applied in a wide range of operations, including assembly, material handling, welding, painting, dispensing, and inspection [2]. Their use continues to expand across the automotive, electronics, aerospace, logistics, and other industries [2,9].

The welding is one of the most popular examples of how automation enhances the quality of processes and the level of operational safety among industrial robotic tasks [2]. Robotic welding guarantees constant control of the trajectory, the high level of repeatability of technological parameters and minimization of the risks of human exposure to heat, fumes, and other dangerous working conditions [2]. Recent research also reveals that contemporary welding robots are increasingly using path planning and trajectory motion as well as smart control techniques to enhance quality of weld, productivity, and flexibility in industrial surroundings [2].

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Robotic spray painting is another important and technologically demanding application area, and it is considered in the present paper as a representative example of a high-precision industrial task [4,7]. Studies have shown that the demand for robotic paint application manipulators has increased due to their potential in terms of stability, quality, repeatability, and overall efficiency [4,5,7]. Compared with manual painting, controlled robot motion provides more uniform paint application and better repeatability [5,6]. In industrial spray painting, product quality and productivity strongly depend on the repeatability of technological parameters such as spray distance, spray angle, overlap ratio, and end-effector speed [5,6,8]. In addition, recent studies have addressed collision-free trajectory generation and optimal robot placement in collaborative robotic painting systems [10,11].

At the same time, spray painting still poses a significant occupational safety risk [12,13]. Paints may contain volatile organic compounds (VOCs) and aerosols, so exposure control remains a mandatory requirement [12,13]. The main hazards are associated with inhalation of paint vapors and contact with the skin, and these risks should be reduced through engineering measures, organizational procedures, and properly selected personal protective equipment (PPE) [12,13]. However, even when protective measures are applied under real working conditions, a residual risk may remain [12,13]. For this reason, robotic painting systems are attractive not only because of process quality and repeatability, but also because they reduce continuous human exposure to hazardous environments [7,12,13].

Robotic manipulators can be controlled in industry in several ways. One of the common methods is PLC-based control, in which the PLC sends commands to the manipulator and determines the sequence of operations [3,14]. PLC-based control is reliable, widely used in real industrial production, and can be easily integrated with sensors, actuators, and safety logic [3]. This helps to maintain process repeatability and deterministic operation [3,14]. However, when a task requires the coordination of many different manipulators, PLC-only control is not always convenient for complex calculations and flexible coordination [18].

Another approach is MATLAB/Simulink-based control, in which the software environment is used to test control algorithms and then send control commands to the manipulator [15-17]. This approach is effective for trajectory planning, model-based control, and data processing, which makes it useful for experimentation and rapid development [15,16]. At the same time, for continuous industrial use, MATLAB-only control may be less practical than PLC-based control,

since it usually depends on a PC environment and is not itself focused on industrial reliability and safety [15,17].

For this reason, this paper employs a combined PLC-MATLAB approach to coordinate several robots within a single automated technological system [18]. The PLC is used for sequence control, interlocks, and device-level coordination, while MATLAB is used for higher-level supervision, trajectory-related processing, and monitoring [3,18]. The PLC and MATLAB communicate through a standard industrial communication interface, so the system maintains industrial stability while also taking advantage of MATLAB computational tools [18].

II. METHODOLOGY

A. TECHNOLOGICAL PROCESS OVERVIEW

Figure 1 illustrates the overall workflow of the automated multi-robot painting line, which highlights the main technological stages from painting to pick and place process. The manufacture process starts when a workpiece is placed on the conveyor line and then, it is transported to the painting station. Once the product reaches the already predefined position, the conveyor belt stops to ensure accurate positioning, with the help of proximity sensors, which are located in each technological stage.

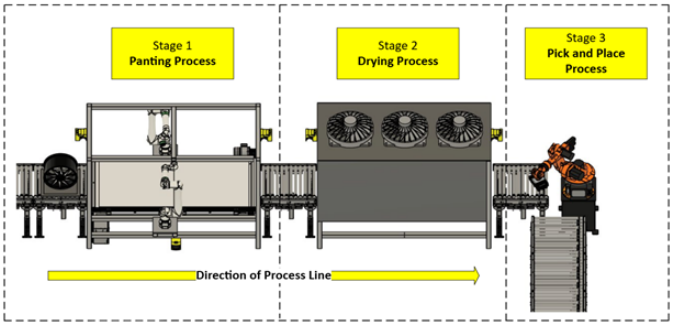


Fig. 1. Structural overview of the automated multi-robot painting process workflow

At the painting station, two RoArm-M2-S manipulators are used to simultaneously apply the coating at the same time in the painting station. Their motions are coordinated by a distributed control layer and switches the spray enable or disable mode.

After the coating operation is completed, the conveyor will transport the workpiece to the drying area. Drying is accomplished with a series of industrial fans which deliver the air flow in several directions uniformly. The drying cycle follows a predefined technological recipe to stabilize the applied coating. When drying is completed, the job is then picked up by the conveyor and continues to move on.

In the final stage, the KUKA manipulator takes the inspected work item off the conveyor and puts it within the specified output platform where finished items are placed. This step completes the technological cycle and prepares the system for processing the next workpiece.

The robots and the conveyor belt are placed at fixed positions in the workspace and do not move between sessions. Because their locations are always the same, no calibration is needed before each run. MATLAB plans all robot movements within a predefined safe area, and any point outside this area is automatically rejected before the motion starts. At each stage of the process, proximity sensors confirm that the workpiece has arrived at the correct position. The PLC only allows the robots to start moving after it receives a confirmation signal from the sensors. This way, the robots and the conveyor belt always work in sync without any additional alignment procedure.

B. HARDWARE & SYSTEM ARCHITECTURE

TThe Hardware Architecture of the automated multi-robot painting line is shown in Figure 2. The system is organized as a unified information and control system that combines control hardware, sensors, actuators, and external analysis tools.

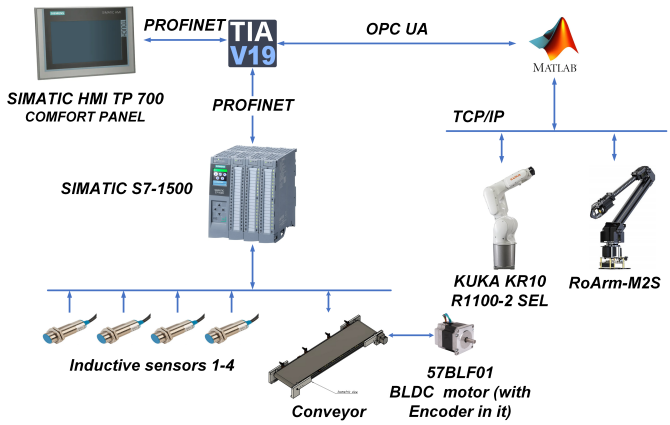


Fig. 2. Structural scheme of the hardware architecture of the multi-robot painting line

MATLAB is a high-level software environment and programming language, that is used for modeling, simulation, numerical computation software and control system design.

TIA Portal is engineering environment, which is used for configuring, programming and controlling industrial automated systems.

OPC UA is Open Platform Communications Unified Architecture. This is an industrial communication standard, which is used for secure and independent data exchange between different devices.

PROFINET is Process Field Network. This is a real-time industrial Ethernet communication protocol, which is used for fast and reliable real-time data exchange between controllers and another device.

HMI is Human Machine Interface. This is a system which allows a supervisor in order to monitor and control a machine through visual interface.

SCADA is Supervisory Control and Data Acquisition. This is a system which is used to monitor and control industrial processes in an advanced mode.

The core of the system is a SIMATIC S7-1500 PLC (CPU 1513F-1 PN), which performs cyclic control, processes

sensor inputs, and controls actuators, the conveyor, and auxiliary equipment. As a failsafe controller, it supports both standard automation and SIL/PL-compliant safety functions, while communication with external systems is implemented via OPC UA and PROFINET.

The hardware architecture of the automated multi-robot painting line is centered on a SIMATIC S7-1500 PLC (CPU 1513F-1 PN), which performs cyclic control, processes sensor inputs, and controls actuators, the conveyor, and auxiliary equipment. As a failsafe controller, it supports both standard automation and SIL/PL-compliant safety functions, while communication with external systems is implemented via OPC UA and PROFINET. Operator interaction is provided through a SIMATIC HMI TP700 Comfort panel, while high-level monitoring, data acquisition, and trajectory-related analysis are implemented in MATLAB/Simulink. At the field level, the system includes inductive proximity sensors for workpiece and conveyor position detection, a BLDC conveyor motor for transport motion, and robotic manipulators: RoArm-M2-S for painting and KUKA KR10 R1100-2 for pick-and-place operations.

The software architecture of the proposed system is organized into four layers, as shown in Figure 3. The mission/task layer is related to overall task control. The motion planning layer is used for waypoint generation, trajectory interpolation, and motion timing. The robot driver layer contains separate control modules for the RoArm-M2-S and KUKA KR10 robots. The communication/transport layer provides data exchange through TCP/IP and for KUKA manipulator via KukaVarProxy. This structure separates planning, robot control, and communication functions within one software framework.

Wireshark was used to monitor network traffic during the experiments. During active robot operation, packet throughput was stable at 20-110 packets per second. A brief burst at the start of each session and some short TCP errors were observed, but no lasting connection failures occurred. This confirms that the OPC UA link between the PLC and MATLAB was reliable throughout all experiments[19].

The SIMATIC S7-1500 PLC monitors joint position data sent from MATLAB via OPC UA. If positions stop changing during active robot motion-indicating a crash or connection loss-the PLC immediately stops the conveyor, disables the spray actuators, and shuts down all robots. Since all network issues observed in the experiments were short and only occurred outside of active motion phases, the 2-second monitoring window provides a sufficient safety margin.

C. SIMULINK MODEL FOR PICK AND PLACE PROCESS

The Simulink model in Figure 3 presents the control architecture used for the KUKA high-DOF manipulator. It integrates the main kinematic and dynamic subsystems, including a Trajectory Generator, Inverse Kinematics, Forward Kinematics, and Inverse Dynamics blocks[15,16,17].

The *Trajectory Generator* computes the desired task-space reference as a homogeneous transformation matrix T. The *Inverse Kinematics* block converts the target pose into the

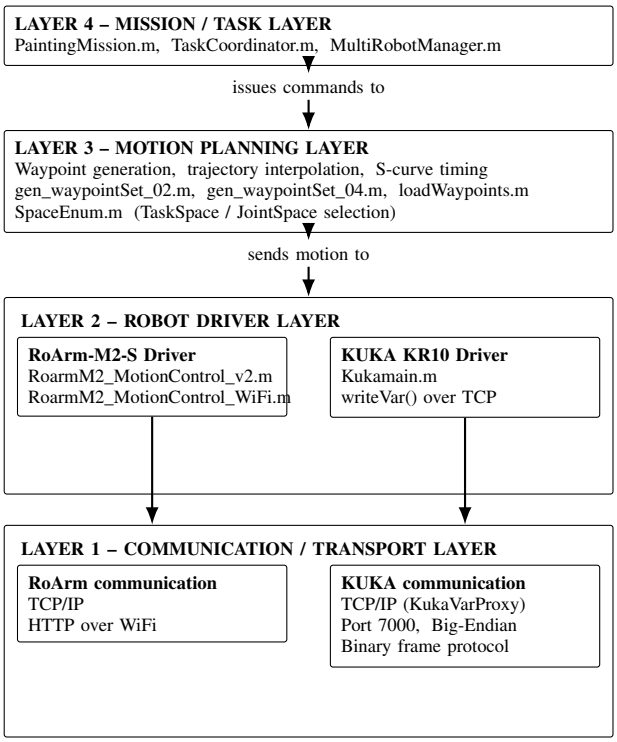


Fig. 3. Structural scheme of the four-layer software architecture of the proposed control framework

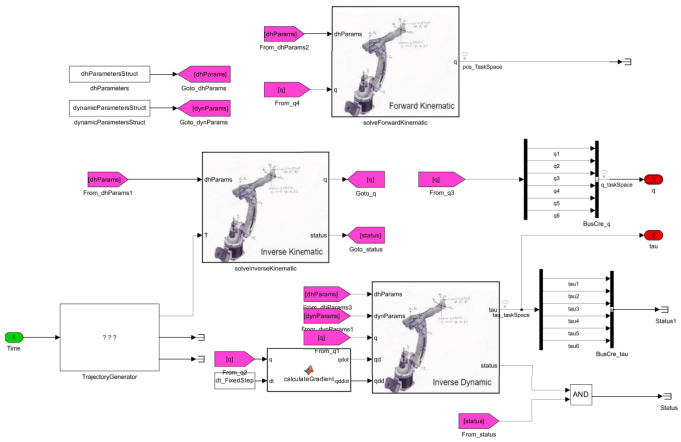


Fig. 4. Simulink model of the KUKA KR10 R1100-2 manipulator control architecture

joint position vector q, which is required to translate Cartesian commands into joint-level inputs [16,17]. The *Forward Kinematics* block verifies the resulting motion by computing the end-effector pose in task space from q, supporting monitoring and feedback. Finally, the *Inverse Dynamics* block estimates the required joint torques τ using q, $\dot{q}$, $\ddot{q}$ and the manipulator model parameters.

The Inverse Dynamics block computes the required joint torques τ using the joint position q, velocity $\dot{q}$, and acceleration $\ddot{q}$. Since $\ddot{q}$ cannot be measured directly, it is obtained in two ways. First, the *TrajectoryGenerator* block uses a quintic polynomial trajectory, so $\ddot{q}$ is obtained directly as the second

derivative of the planned trajectory; therefore, no additional estimation is required. Second, the *calculateGradient* block, visible at the bottom of Figure 4, computes $\ddot{q}$ by numerically differentiating the joint position signal using a fixed time step dt.

It should be noted that numerical differentiation is suitable in simulation, where signals are smooth and noise-free. In real hardware implementations, however, this approach would amplify sensor noise, and a dedicated state observer or filtering method would be required instead.

Each joint servo is modeled as a second-order closed-loop transfer function:

$$\frac{Q_i(s)}{Q_{i,\mathrm{ref}}(s)} = \frac{\omega_{n,i}^2}{s^2 + 2\zeta_i\omega_{n,i}s + \omega_{n,i}^2} \tag{1}$$

where $\omega_{n,i}$ is the natural frequency and ζ_i is the damping ratio of the i-th joint servo. This model was used to represent the closed-loop servo dynamics of each joint during trajectory generation.

D. TASK EXECUTION & CONTROL LOGIC

The main controller of the system is a PLC (programmable logic controller) and is responsible for the main control logic. The main program was developed in the Ladder logic (LAD), which is an electrical logic language widely used in programming. The control code was organized using a block diagram as shown in Figure 5, where each function is controlled in a simple and modular way.

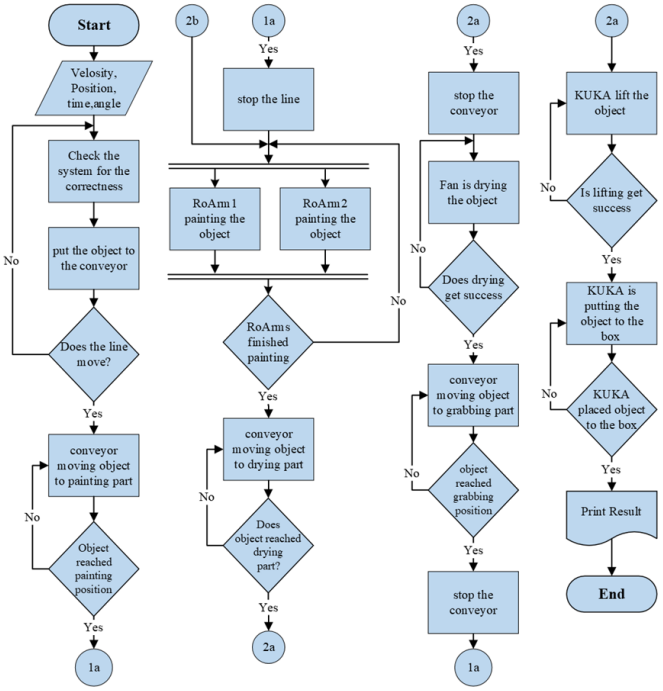


Fig. 5. Block diagram of the automated multi-robot process control sequence

This structure simplifies the diagnosis and monitoring of the system, since each stage of the process is performed in a certain sequence. If an error occurs, it is easier to determine at what stage the sequence of actions was interrupted and which condition was not met.

The RoArm-M2-S manipulators and MATLAB operate within the same subnet and are connected to a common router, enabling data exchange via the TCP/IP protocol. In the painting subsystem, MATLAB transmits motion commands to the RoArm-M2-S manipulators in JSON format. Figure 6 shows the 3D preview of path planning motion. Based on the received target coordinates, the manipulators perform trajectory planning and execution using the MATLAB framework, ensuring efficient and collision-free motion.

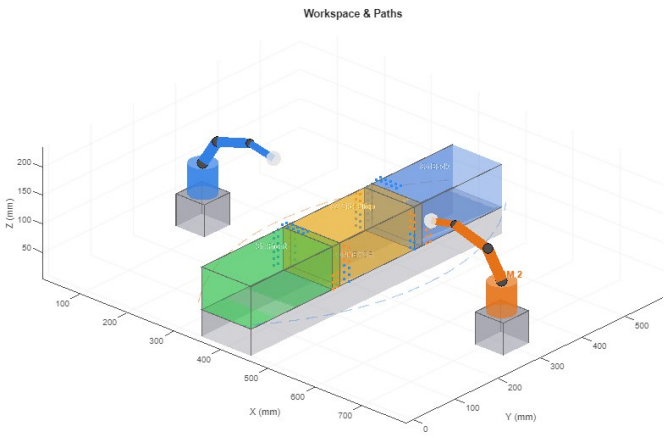


Fig. 6. Three-dimensional workspace and trajectory path preview of the RoArm-M2-S manipulator

The KUKA KR10 R1100-2 robot is also connected to MATLAB within the same subnet, which enables TCP/IP communication. Control data are transmitted from MATLAB using the KUKAVARPROXY message format. Figure 7 shows the 3D preview of path planning motion. The proxy writes the received values into global KRL variables, which are accessed by the KUKA control program to execute motion commands or trigger corresponding control logic.

MATLAB additionally receives real-time feedback data from the servo drives of the manipulators. These data include motor torque and joint position signals and are visualized using real-time plots. This feedback enables online monitoring and performance analysis of the robotic system during operation.

III. RESULTS

The experimental evaluation confirmed that the proposed dual-arm painting system was able to execute the surface coating process in a coordinated, stable, and repeatable manner. The real-time telemetry recorded from both RoArm manipulators, presented in Figures 6, shows that the generated Cartesian trajectories accurately follow the intended raster-based painting pattern. The periodic variation of the position curves corresponds to the sequential execution of approach, surface sweep, and transition motions between adjacent coating stripes. Moreover, the gradual displacement of the trajectory coordinates reflects the systematic progression

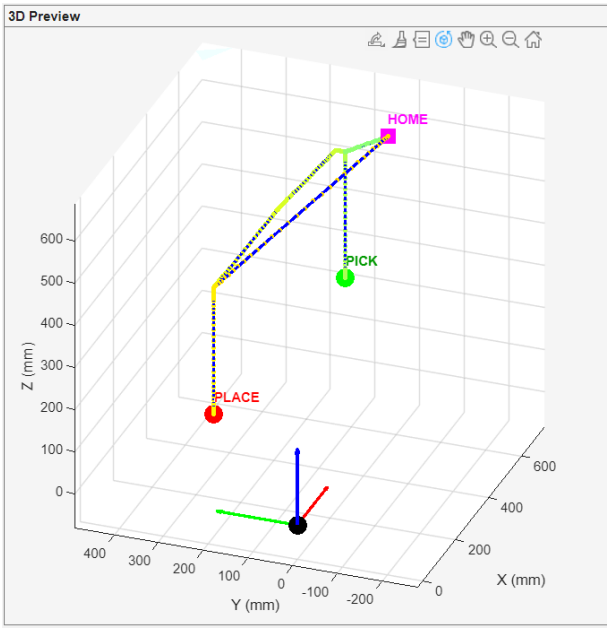


Fig. 7. Three-dimensional trajectory path planning preview of the KUKA KR10 R1100-2 manipulator

of the manipulators across the front, side, top, and back surfaces of the workpiece during the painting process.

The estimated joint torque profiles shown in the same figures further indicate stable mechanical loading throughout the painting cycle. The highest torque values are observed in the shoulder and elbow joints, which is consistent with the kinematic structure of the robotic arms and the positioning requirements of the custom spray end-effector during surface coverage. No critical overloads or unstable dynamic behaviour were detected during the experiments, and the torque levels remained within the expected operating range of the RoArm platform. Overall, these results confirm that the proposed control architecture provides reliable dual-arm coordination for robotic painting while also supporting real-time monitoring and further performance analysis.

The results confirm stable OPC UA communication between PLC and MATLAB, successful execution of synchronized RoArm routines, and reliable real-time position monitoring[19].

In Figure 7 the KUKA KR10 R1100-2 manipulator is used for pick-and-place. This process trajectory was generated and used in the experiment, which is depicted in Figure 8. The path has 190 Cartesian waypoints and is found in three views namely 3D, XY plane top view, and XZ plane side view. The 3D visualization indicates a linear and continuous displacement between the pick and place position. The color change on the path indicates the sequence of the waypoints. The starting and the ending points are identified and noticeable [16]. The operation process consists of the key stages: approach, lifting, transfer, and placement.

The XY plane indicates that there is a straight and planned horizontal flow between the two positions. This implies that the robot does not move around unnecessarily and only

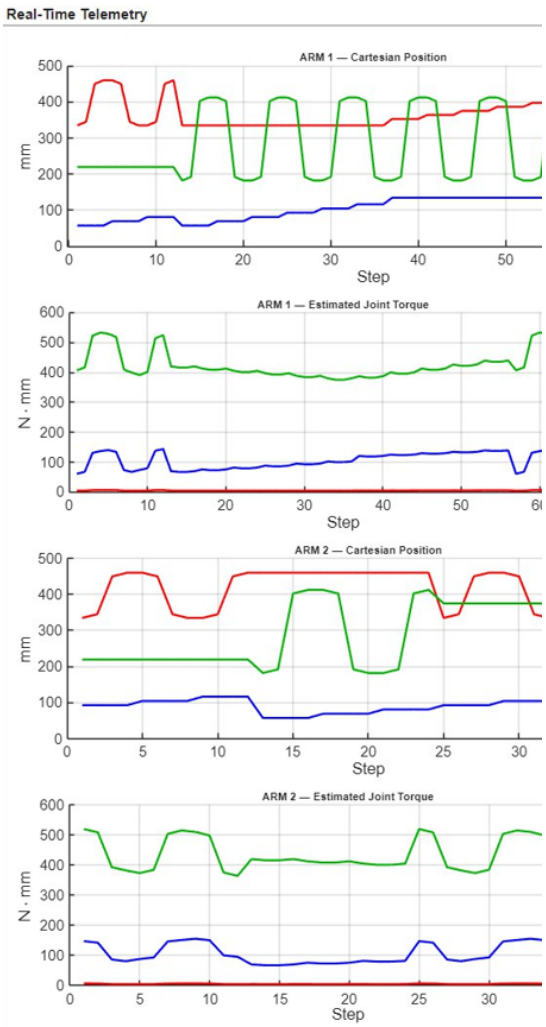


Fig. 8. Real-time telemetry of the painting process: Cartesian position and estimated joint torque for both RoArm-M2-S manipulators

travels between the pick and place zones. The side view (XZ plane) indicates an upward lifting action at the time of picking of the object and a downward controlled action at the time of placement. This vertical movement gives the safe clearance over the working area. The findings validate that the resulting path is continuous, stable, and can be used in an industrial robotic system in a pick-and-place operation which can be executed reliably.

The real-time motion data of the RoArm-M2-S manipulators and the KUKA robot were used to ensure that both systems followed the established trajectories and continuously reached the set target positions. This reflects improved overall accuracy and helps to improve the safety of the painting and pick-and-place operations.

In Figure 9, the pick-and-place experiment also recorded real-time telemetry of the KUKA KR10 R1100-2 manipulator. The data that was monitored was Cartesian position, joint torque, joint velocity, and motor current. These data were employed to monitor the robot movement in the process of executing the generated trajectory and to compare the

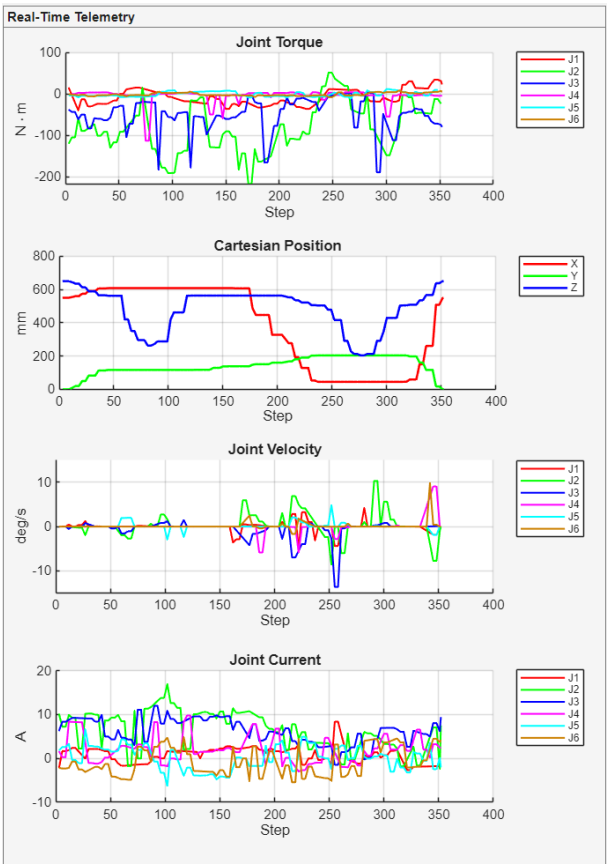


Fig. 9. Real-time telemetry of the pick-and-place process: joint torque, Cartesian position, joint velocity, and motor current for the KUKA

measured behaviour with the commanded path. The received plots demonstrate the variations of the robot position and joint-level parameters at various points of the pick-and-place cycle, such as approach, lifting, transfer, and placement. Therefore, the telemetry data offered further experimental evidence that the KUKA manipulator was able to follow the generated trajectory in real time.

IV. CONCLUSION

This paper introduces a PLC-based control system of the application of an industrial paint line consisting of a multi-robot system and external devices into one automated system. It utilized the Siemens S7-1500 PLC as the core layer of implementation of deterministic sequence, lockout, and device-level coordination, and the MATLAB control layer as the coordination and monitoring layer using the standardized OPC UA data exchange. The creation of the implementation showed that it provides reliable delivery of commands, results in consistent synchronization of states between the PLC and MATLAB, and consistent collection of feedback data about robot movement in order to monitor and analyze it.

The following are some of such boundaries that could be enhanced. In the present implementation it was found that the movement of the robot is largely controlled by predefined logic and trajectories that do not allow adapting to changes in the environment, part position or process conditions. Our next project, as a working step, will be the use of a decision making layer of a reinforcement learning (RL) to enhance the precision of movement and process repeatability by learning to optimize the parameters of trajectory like speed, spray distance and trajectory correction with the help of real time feedbacks..

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